The Political Economy of Clientelism: When Inequality Favors the Elites

Syahid Izzulhaq
Diponegoro University
Akhmad Syakir Kurnia (akhmadkurnia@lecturer.undip.ac.id)
Diponegoro University

Research Article

Keywords: Clientelism, Democracy, Income Inequality, Collective-action

Posted Date: October 12th, 2023

DOI: https://doi.org/10.21203/rs.3.rs-3424520/v1

Additional Declarations: No competing interests reported.

The Political Economy of Clientelism: When Inequality Favors the Elites

Syahid Izzulhaq[1]
https://orcid.org/**0000-0001-8037-0105**

Akhmad Syakir Kurnia[1a]
https://orcid.org/**0000-0002-1457-2101**

[1] Department of Economics and Development Studies, Diponegoro University, Semarang, Indonesia.

[a] Corresponding Author:

Akhmad Syakir Kurnia
Building C, FEB Tembalang Campus,
The Faculty of Economics and Business,
Diponegoro University,
H. Prof. Soedarto SH. Street,
50275 Tembalang,
Semarang, Indonesia.
Email: akhmadkurnia@lecturer.undip.ac.id

The Political Economy of Clientelism: When Inequality Favors the Elites

Abstract

This paper provides a novel foundation of inequality and political clientelism. First, we conduct an extensive-form game analysis to illustrate the nexus. Our paper demonstrates that with a high concentration of wealth, implying a lower income of people experiencing poverty, the elites can conduct political clientelism since it is enforceable and effective to capture sufficient de facto and collective-action power. We then conduct empirical investigations using two-stage least-squared estimations based on 104 democratic countries from 2002 to 2017 to test our theoretical conjecture. The empirical results suggest a strong correlation between clientelism and income inequality, confirming our anticipated pattern.

Keywords: Clientelism, Democracy, Income Inequality, Collective-action.
JEL Classification: P16, H23, D63.

1 Introduction

Studies in political clientelism have garnered significant scholarly attention, yet it is likely to remain in its early phases of development. Berenschot (2018) emphasized that the existing literature lacks intense discussions on the link between economic outcomes and clientelistic practices. At least two well-established perspectives have addressed the link between economic growth and political clientelism: Cost-based and constraint-based perspectives. The cost-based view argues that economic development should necessarily limit political clientelism. Rising incomes and the middle class's growth reduce the yields and increase the cost of politicians' clientelistic strategies. Magolini, Diaz-Cayeros, and Estévez (2007) argued that as a country develops, clientelism should naturally be eroded as it costs the ruling elites extensively. Conversely, the constraint-based perspective claims that the clientelistic strategy would be ineffective with a relatively low economic source concentration. It allows for a more robust public sphere and practical disciplining effect of politico-business elites (Berenschot 2018).

While a growing literature on the political economy of clientelism considers cost and constraint perspectives differently, we argue that these perspectives are neither inconsistent nor refuting each other, while it virtually provides a holistic view of the inequality-clientelism nexus. The cost-based perspective gives an intuition about the cost feasibility of political clientelism under particular economic conditions. The constraint-based perspective provides the importance of collective action in curtailing political clientelism.

This paper provides a novel foundation of inequality-clientelism nexus incorporating cost and constraint perspectives. First, we conduct an extensive game-theoretical analysis to capture the link between political clientelism and economic development in the form of income inequality. We then perform an empirical test asking whether or not our theoretical conjecture is valid. The test uses unbalanced panel data, providing cross-country evidence of the inequality-clientelism nexus. Such investigation is crucial as, surprisingly, systematic comparative studies on clientelism are limited (Berenschot 2018).

We develop the theoretical model roots to Acemoglu and Robinson (2000, 2008) with several alterations provided our focuses. Acemoglu and Robinson (2008) demonstrated a rigorous theoretical model examining the elites' investment to maintain their de facto political significance in a democratic regime so that the elites could take control over economic policy. We use this intuition to illustrate the link between clientelistic strategy, de facto political power, and economic policy. We then extend the model to explain the link between political clientelism and de facto political power and the power of collective actions. While their work mentioned no collective action role, it is a crucial feature in democracy (Bellinger and Arce 2011; Moseley 2015). We model the collective action in protest and replicate Acemoglu and Robinson's seminal work. They demonstrated that the elites should choose to franchise their political power or maintain their political significance by promising private benefits to the non-elites. However, the elites would only promise personal benefits, which is uncertain for the non-elites. It then allows for pivotal protests by the non-elites to force the elites to franchise their political power. Unlike Acemoglu and Robinson (2000), we consider no uncertainty about the elites' transfer. In a democratic regime, a clientelistic strategy means that the ruling elites provide promises and personal benefits for political support. To some extent, therefore, political clientelism could lead to serious collective action problems.

Our model illustrates that income inequality is central to clientelistic practices. When income inequality is sufficiently high, it is rational for the elites to conduct political clientelism. In this case, the elites are privileged to utilize their wealth to increase de facto power by providing private benefits for some fraction of the non-elites in exchange for political support. Consequently, it captures the ability to construct a protest against the elites, leading to collective-action problems. In other words, by obtaining sufficient de facto power, the elites can weaken the power of protest, adversely impacting the public sphere and pivotal control over the ruling elites' actions.

On the contrary, clientelistic strategies would be unreasonable for the elites when income inequality is sufficiently low. Adjacent to the cost-based perspective, our model illustrates that the more equitable the income with low-income voters getting better off, clientelism should diminish since clientelism is costly for the ruling elites. The model suggests that aligning with the constrained-based perspective, when the concentration of economic sources, i.e., wealth, is sufficiently low, the clientelistic strategy would be ineffective in lessening the protest power. In this circumstance, the elites would rationally conduct no clientelistic strategy as income equality implies an escalating robust public sphere with a compelling political check and balance mechanism.

Income redistribution often takes an inefficient form due to the political commitment problem regarding employment incentives in the public sector (Robinson and Verdier, 2013). Public sector employment is selective and reversible, providing sustaining utility for a voter in exchange for political support to a particular politician. Hence, clientelistic practices become more attractive in a situation of high inequality and low productivity. Our paper develops a model based on Robinson and Verdier (2013) by distinguishing each voter group's de jure and de facto power. In a democracy, de facto control often deviates from the de jure power. The de jure authority is associated with the power that ensures each individual has equal political rights. It does not regulate how each individual utilizes their political rights. Poor groups would like to use their de jure power to support the elites' political interest, increasing their de facto power. As the wealthy group, the ruling elites use their wealth to grip more considerable de facto control by providing private benefits for some fraction of the non-elites in exchange for political support (Berenschot 2018). Such a mechanism is effective as poor voters are risk-averse and more easily tempted by clientelistic benefits (Brusco, Nazareno, and Stokes 2004; Scott 1972). We extend our model by explicitly incorporating collective action issues in a protest movement and putting the relationship between political clientelism and the power of protest into scrutiny. The incorporation of protest power is noteworthy as a protest is a form of public sphere openness that prevents abusive elitist rule (Bellinger and Arce 2011; Moseley 2015).

This paper conducts an empirical investigation to test our theoretical conjecture. We perform instrumental variables panel data regressions based on a dataset covering, in total, 104 democratic countries from 2002 to 2017. From the investigation, our empirical results confirm the theoretical conjecture, suggesting that clientelism is strongly related to income inequality in the way we predict. Moreover, as income inequality rises, political clientelism will be more pervasive. As our model illustrates, a high concentration of wealth, implying a lower income of low-income people, allows the elites to conduct political clientelism because it is enforceable and effective to capture sufficiently de facto and collective action power. This finding is also consistent in various robustness exercises.

The rest of this paper is organized as follows: Section 2 elaborates on the basic model that comprises the economic structure, the link between political clientelism and elites' de facto power, and its consequences on collective action, redistribution policy, and income inequality. Section 3 provides a simple game-theoretical analysis to illustrate the strategic interaction between the elites and non-elites. Section 4 discusses the shutdown of political clientelism. Empirical strategy and empirical discussions will be provided in Sections 5 and 6, respectively. Finally, section 7 elaborates on robustness exercises, and concluding remarks are provided in section 8.

2 The Baseline Model

2. 1 Economic Structure, Democracy, and Moderating Effect

In this section, we replicate a hypothetical economic structure following Acemoglu et al. (2015) and Acemoglu and Robinson (2006) to illustrate the moderating effect of democracy (Meltzer and Richard 1983). We consider that the economy consists of n population divided into two groups, namely elites (r) and non-elites (p), $i = (r, p)$. The average income of all groups is expressed as follows:

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y^i \tag{1}$$

where $\bar{y}$ is the average income; y^i is pre-tax income (i.e., the total income of agent i when a tax, τ, and transfer has not functioned yet, $y^i(\tau = 0)$). We normalize the population n by $n = 1$ where the number of the non-elite group exceeds the elite group that is $n_p > n_r$, where $(1-\delta) = n_p;\ n_r = \delta$; therefore, $n = \delta + (1-\delta) = 1$.

We now consider that the government runs a lump-sum redistribution policy by levying the tax at τ rate. Therefore, the government budget constraint takes form as follows.

$$T \leq \tau\bar{y} - C(\tau)\bar{y} \Rightarrow T \leq \big(\tau - C(\tau)\big)\bar{y} \tag{2}$$

where $C(\tau)\bar{y}$ is the distortionary cost of taxation that is assumed to be $C'(0) = 0$, $C'(1) = 1$, $C'(\cdot) > 0$, $C''(\cdot) > 0$, and the tax rate is τ: [0,1].

We proceed by defining the i's post-tax income as the following equation.

$$\hat{y}^i(\tau) = y^i - \tau y^i + T \Rightarrow \hat{y}^i(\tau) = (1-\tau)y^i + \big(\tau - C(\tau)\big)\bar{y} \tag{3}$$

Equation (3) implies that the i's post-tax income is the sum of the pre-tax income and transfer minus the fraction τ of the i's pre-tax income defined as follows.

$$y^p = \frac{(1-\theta)\bar{y}}{(1-\delta)} \text{ and } y^r = \frac{(\theta)\bar{y}}{(\delta)} \tag{4}$$

In this case, therefore, the post-tax income would be equal to pre-tax income when there is no fraction of i's income imposed by the tax; therefore, $y^p = \hat{y}^p(\tau = 0)$ and $y^r = \hat{y}^r(\tau = 0)$. Now, we proceed by defining θ as a portion of $\bar{y}$ concentrated in the elites. In other words, θ represents income inequality.

Our analysis's baseline structure considers that the initial distribution of wealth among agents is concentrated in the elites. Simultaneously, the non-elites possess less wealth, in a portion larger than the elites, $\theta > \delta$. In this case, therefore, the pre-tax income structure for each agent is expressed as follows.

$$y^p < \bar{y} < y^r \tag{5}$$

Equation (5) explains that the elites are a small group with a more substantial income portion of the economy than the non-elites.

In a democracy, the median voters (M) are the main determinants of the tax rate (Meltzer and Richard 1983). In this case, where the non-elite group is larger than the elites, $(1-\delta) > \delta$ with $y^p < \bar{y} < y^r$, the non-elites thus are the median voters because $(p = M)$. In this circumstance, the tax rate thus sets to which the extent that it maximizes the amount of the median voters' post-tax income (see Acemoglu and Robinson 2006):

$$\frac{y^p}{\bar{y}} = 1 - C'(\tau^p) \text{ or } \tau^p > 0 \tag{6}$$

Equation (6) implies that in a democratic regime, the median voters (i.e., the non-elites) would strictly prefer to set the tax rate above zero to gain the benefits from income redistribution through transfer. In other words, the post-tax income of the non-elites is increasing when it is under a democratic regime, while the elites' post-tax income would be lower than their pre-tax income since $\bar{y}$ is constant. As such, income inequality would be reduced by democracy.

Proposition 1. Under a democratic regime, the non-elites are the decisive voters $(p = M)$ and set the optimum tax rate by $\tau = \tau^p > 0$.

Proposition 1 expresses the moderating effect of democracy where the post-tax income of the non-elites is higher than the pre-tax income, $y^p < \hat{y}^p(\tau = \tau^p)$, while the elites' post-tax income is lower than the pre-tax income, $\hat{y}^r(\tau = \tau^r) > y^r$. Consequently, it lowers income inequality, $\hat{\theta}(\tau = \tau^p) < \theta$.

2. 2 Clientelism and *De Facto* Political Power

Democracy ensures that de jure political power is equally distributed among individuals. In this case, de jure power possessed by the non-elites thus exceeds the elites' de jure power, given that the population of non-elites is larger than the elites, $n_p > n_r$. However, de jure power ensures everyone has equal political rights but does not regulate how each individual utilizes it, so there might be a deviation between de jure and de facto power. For instance, some fraction of the non-elites might use their de jure power to support the elites' political interest, increasing their de facto power. On the contrary, each individual in the group of non-elites also utilizes their political power on behalf of their group's interest so that the non-elites de jure control equals its de facto power. In this case, democracy would virtually benefit the non-elites at the elites' expense, where the political equilibrium would be stated precisely as in Proposition 1.

Thus, it is reasonable for elites to maintain their political significance through de facto power. In this case, the elites can utilize their wealth to increase de facto control by providing non-policy-based private benefits for some fraction of the non-elites in exchange for political support (Berenschot 2018). This practice is well-known as political clientelism, an effective strategy to gain political support. Although democracy possibly makes clientelism less exploitative, it is not necessarily less pervasive (Chandra 2004; Okthariza 2020; Berenschot 2018; Aspinall 2013; Aspinall and Berenschot 2019). Brusco, Nazareno, and Stokes (2004) and Scott (1972) argued that clientelistic strategies would work better since poor voters tend to be risk-averse and, hence, more easily tempted to patron-client relationship benefits.

Suppose the following expression shows the de facto power functions possessed by the elites and non-elites:

$$\begin{aligned} \varphi_r(\beta) &= \delta + \vartheta(\beta)(1-\delta) \\ \varphi_p(\beta) &= (1-\delta) - \vartheta(\beta)(1-\delta) \end{aligned} \tag{7}$$

For $\varphi_r + \varphi_p = 1$.

where φ_r and φ_p respectively denote de facto power possessed by the elites and non-elites; β: $[0,1]$ is some portion of the elites' income transferred to a fraction of non-elites population, $\vartheta(\beta)$, which represents the degree of political clientelism. Also, $\vartheta(\beta)$ is assumed to be $\vartheta(0) = 0$; non-decreasing in the level, $\vartheta'(\cdot) > 0$, which means that $\vartheta(\beta)$ rises by the increasing β; and $\vartheta(\beta)$ increases faster as β increases.

De facto power then plays a crucial role in determining economic policy. The tax rate and transfer would be determined by comparative de facto power possessed by the elites and non-elites. By following Acemoglu and Robinson (2008), we presume that the tax rate would be determined based on the following conditions:

$$\tau^i = \begin{cases} \tau^r \text{ if } \varphi_r(\beta) > \varphi_p(\beta) \\ \tau^p \text{ if } \varphi_r(\beta) < \varphi_p(\beta) \end{cases} \tag{8}$$

Equation (8) explains that when $\beta = 0$, then $\varphi_r(\beta) < \varphi_p(\beta)$; so that the tax rate is favorable for the non-elites, τ^p. In other words, it illustrates that with no political clientelism, the non-elites would possess equal power between de facto and de jure, where $\varphi_p = n_p$ so that $\varphi_r < \varphi_p$. As such, democracy would generate lower income inequality, as expressed in Proposition 1. On the contrary, by defining that β^* satisfies $\varphi_r(\beta^*) = \varphi_p(\beta^*)$ and the elites conduct $\beta_r > \beta^*$; hence, the elites would gain a higher de facto power than the non-elites, $\varphi_r(\beta_r) > \varphi_p(\beta_r)$. In this situation, the tax rate would be τ^r.

Proposition 2. If and only if the elites transfer β_r portion of their income in exchange for the non-elites political support for $\beta^* < \beta_r$, the elites would sufficiently possess $\varphi_r(\beta_r)$ de facto power and ability to set the tax rate that optimizes their post-tax income, $\tau = \tau^r$ because $\varphi_r(\beta_r) > \varphi_p(\beta_r)$.

2. 3 Clientelism and the Power of Protest

In a democracy, public sphere openness in protest is essential as a contrarian against abusive elitist rule (Bellinger and Arce 2011; Moseley 2015). However, whether or not protests could effectively gain enough power is conflicting. In our framework, we encompass the importance of protest in a democratic regime and emphasize the impact of political clientelism on the power of protest. We argue that political clientelism captures the political power of the non-elites and the ability to construct a protest against the elites. In this regard, political clientelism would mobilize some fraction of the non-elites, raising the collective-action problems to oppose the elites' action, stifling the public sphere, and ineffective scrutiny and disciplining affect politico-business elites (Berenschot 2018). In support of this argument, Kurtz (2004) also showed that there had been a substantial decline in the capacity of social actors to organize and mobilize a protest due to the collective action problem posed by, in particular, free-market contexts that oppose intervention in income redistribution policy.

The protest succeeds when it satisfies many participating people (Barbera and Jackson 2016). Suppose that the non-elites need to gather an amount of $\aleph \geq \aleph^p$ protesters to realize the protest. $\aleph^p$ is the minimum amount of the protesters to succeeding the protest where, intuitively, the protest would succeed if the non-elites possessed de facto power equally or more in compared to the elites, $\varphi_p = \varphi_r$, so that it requires a minimum of $n/2$ protesters. Furthermore, $\aleph$ is the actual participation of the non-elites in the protest, which is defined as follows:

$$\aleph = (1-\delta) - \vartheta(\beta)(1-\delta) \Rightarrow \aleph = [1-\vartheta(\beta)](1-\delta) \tag{9}$$

where $(1-\delta)$ is the total population of non-elites, and $\vartheta(\beta)(1-\delta)$ is the number of non-elites that refuse to join the protest as they are mobilized by the elites.

We now consider that β^* satisfies $\aleph^* = \aleph^p$ where $\aleph^* = [1 - \vartheta(\beta^*)](1-\delta)$. In other words, the protest would have success if the elites provide $\beta_{\Xi} \leq \beta^*$ of their income as it satisfies $\aleph \geq \aleph^p$. Otherwise, when the elites decide to provide $\beta_{\beth} > \beta^*$ for clientelistic strategy, it leads to the insufficient number of protesters, $\aleph < \aleph^p$; therefore, the protest would fail. Intuitively, when the elites provide $\beta_{\Xi} \leq \beta^*$ portion of their income in exchange for political support, the non-elites would be better off joining the protest. It thus leads to a sufficient number of protesters. On the contrary, the non-elites would be better off if they refused to join the protest when the elites only provide $\beta_{\beth} > \beta^*$.

Proposition 3. If the elites decide to conduct a clientelistic strategy at $\beta_{\Xi} \leq \beta^*$ where β: [0,1]; and $\aleph^{\Xi} \geq \aleph^p$; so that the protest would succeed. If the elites agree for $\beta_{\beth} > \beta^*$, where β: [0,1]; and it thus leads to an insufficient number of the protesters, $\aleph^{\beth} < \aleph^p$, and, therefore, the protest would fail.

Proof: See Appendix A.

2. 4 Rational Constraint of Clientelism

This section aims to clarify the rational space of the elites' clientelistic practices. It is important to discuss because the elites should have a boundary to conduct a clientelistic strategy, by which, to some extent of β, it could be either too expensive or ineffective for the elites. Firstly, we define the upper-bound of the elites' investment, $\overline{\beta}$, as a condition when *de facto* investment satisfies an equal amount of the elites' net income in captured democracy and their post-tax income in democracy:

$$\hat{y}^r(\tau = \tau^r) - \overline{\beta} y^r = \hat{y}^r(\tau = \tau^p) \Rightarrow \overline{\beta} = \frac{\hat{y}^r(\tau = \tau^r) - \hat{y}^r(\tau = \tau^p)}{y^r}$$

Recall that $y^r = \theta\bar{y}/\delta = \hat{y}^r(\tau = \tau^r)$,

$$\overline{\beta} = 1 - \delta \frac{\hat{y}^r(\tau = \tau^p)}{\theta \bar{y}} \tag{10}$$

where $0 < \overline{\beta} \leq 1$ because $0 < \delta \frac{\hat{y}^r(\tau=\tau^p)}{\theta\bar{y}} \leq 1$ for $\hat{y}^r(\tau = \tau^p) < \theta\bar{y}$.

Now, we proceed by defining the lower bound of the elites' clientelistic strategy, β^*, that satisfies a condition where the non-elites benefit by becoming protesters and non-protesters is indifferent, that is $y^p + \beta^* y^r/(1-\delta) = y^p + ((1-\mu)y^r/(1-\delta)) - \varepsilon\bar{y}$. Intuitively, by only providing β^*, the elites would fail to induce collective-action problems, resulting in a more substantial power of protest. As such, we define β^* as the following:

$$\beta^* = (1-\mu) - \frac{\varepsilon(1-\delta)\delta}{\theta} \tag{11}$$

Assumption 1. $\theta > \theta^*$ where $\theta^* = \frac{\hat{y}^r(\tau=\tau^p)}{\mu\bar{y}} - \frac{\varepsilon(1-\delta)\bar{y}}{\mu\bar{y}}$; so that it is always the case that $\beta^* < \overline{\beta}$.

Combining propositions 2 and 3 with the restriction of Assumption 1, we could reasonably arrange the rational space of elites' investment, β, formulated as follows.

$$0 < \beta_{\Xi} \leq \beta^* < \beta_r < \overline{\beta} \leq \beta_{\omega} \leq 1 \tag{12}$$

where β_r is the rational space of elites' investment because β_r satisfies a crucial condition in which $\beta^* < \beta_r < \overline{\beta}$. There will always be a rational space in whole conditions as long as assumption 1 holds.

On the one hand, β_{Ξ} and β_{ω} are irrational spaces of the elites' de facto investment. The main reasons are irrational are that, firstly, the protest would always be successful when the elites merely investing β_{Ξ} fraction of the elites' income; therefore, the outcome of no investment and β_{Ξ} is simply indifferent. In this case, the elites' rational actions would invest β_r or not in order to make their outcome better with investment. Second, investing β_{ω} portion of their income is not enforceable for them as it is too expensive for the elites in terms of the elites' payoff in a stable democracy (i.e., $\hat{y}^r(\tau = \tau^p)$) is weakly larger than in a captured democracy (i.e., $\hat{y}^r(\tau = \tau^r) - \overline{\beta} y^r$).

3 Analysis of the Baseline Model: An Extensive-form Game Approach

Let us now illustrate a perfect information extensive-form game between the elites and non-elites, $i \in (r, p)$. In this game, we presume that the elites would be the first mover, while the non-elites would be the second-move agent that responds to the elites' decisions, as in Acemoglu and Robinson (2000). We define the possible actions taken by the elites as follows.

$$\chi_r = \langle 0, \beta_r \rangle \tag{13}$$

Where β: [0,1] with $\beta_{\Xi} \leq \beta^* < \beta_r < \overline{\beta} \leq \beta_{\omega}$.

Equation (13) expresses that the elites would choose to either invest β_r or zero in the first-choice node. As we discussed earlier, β_r is rational for the elites to ensure sufficient *de facto* power (see equation 14). However, zero investment in *de facto* power is also rational for the non-elites because the payoffs of β_{Ξ}, and $\beta = 0$ is indifferent; as such, the elites would choose no clientelistic strategy rather than isolating β_{Ξ} fraction of their income.

Once the elites decided, the non-elites would protest or accept the situation. Therefore, we define the non-elites possible actions as follows:

$$A_p = \langle \mathcal{P}, \mathcal{A} \rangle \tag{14}$$

Where $\mathcal{P}$ and $\mathcal{A}$ respectively represent "protest" and "accept."

In this regard, the protest would address income redistribution; therefore, its primary objective would be to expropriate some fraction of the elites' income and redistribute it to the non-elites. However, the protest's success depends on the degree of political clientelism β (see proposition 3).

Given these possible actions, the game would end in four possible terminal nodes (Z): $Z_1 = \left(U_1^r, U_1^p\right)$, $Z_2 = \left(U_2^r, U_2^p\right)$, $Z_3 = \left(U_3^r, U_3^p\right)$, and $Z_4 = \left(U_4^r, U_4^p\right)$. The game would end up in $Z_1 = \left(U_1^r, U_1^p\right)$ when the elites choose not to conduct a clientelistic strategy, and the non-elites choose to set up a protest. Under this circumstance, the protest initiated by the non-elites would be a success because the elites are conducting no clientelistic strategy. However, the elites would receive nothing when the protest succeeds (Acemoglu and Robinson, 2000, 2006). Therefore, the elites' and the non-elites' payoff is defined as follows.

$$U_1^r(0, \mathcal{P}) = 0$$
$$U_1^p(0, \mathcal{P}) = y^p + \frac{(1-\mu)y^r}{(1-\delta)} - \varepsilon\bar{y} \tag{15}$$

We now proceed to the situation when the elites choose not to conduct a clientelistic strategy, and the non-elites decide to accept it; therefore, the game ends in $Z_2 = \left(U_2^r, U_2^p\right)$. In this circumstance, *de facto* power would be distributed equally among the citizens as the de jure power; therefore, the tax rate and transfer would be determined by the non-elites as the median voters. Thus, we come up with a stable democracy where the payoff for the elites and non-elites is expressed as follows:

$$U_2^r(0, \mathcal{A}) = \hat{y}^r(\tau = \tau^p)$$
$$U_2^p(0, \mathcal{A}) = \hat{y}^p(\tau = \tau^p) \tag{16}$$

The third terminal node, $Z_3 = \left(U_3^r, U_3^p\right)$, occurs when the elites decide to conduct β_r, while the non-elites choose to run a protest. Under this circumstance, the elites could set the tax rate to the extent that optimizes their payoff (i.e., $\tau = \tau^r$) and simultaneously ensure that the protest initiated by the non-elites would fail (see proposition 2 and 3). Hence, the payoffs are defined as follows.

$$U_3^r(\beta_r, \mathcal{P}) = \hat{y}^r(\tau = \tau^r) - \beta_r y^r$$
$$U_3^p(\beta_r, \mathcal{P}) = \hat{y}^p(\tau = \tau^r) \tag{17}$$

Lastly, the game ends at the fourth terminal node, $Z_4 = \left(U_4^r, U_4^p\right)$, when the elites choose to invest β_r fraction of their income in exchange for non-elites political support, while the non-elites choose to accept the situation. Under this circumstance, the non-elites would receive an extra income from the non-elites, although the tax rate is arranged to optimize the elites' payoff. Therefore, the elites and the non-elites would perceive the following payoffs.

$$U_4^r(\beta_r, \mathcal{A}) = \hat{y}^r(\tau = \tau^r) - \beta_r y^r$$
$$U_4^p(\beta_r, \mathcal{A}) = \hat{y}^p(\tau = \tau^r) + \frac{\beta_r y^r}{(1-\delta)} \tag{18}$$

Let us introduce one condition to ensure the game is systematically formed: the redistribution benefit in a democracy is higher than the additional benefit of a successful protest for the non-elites. Intuitively, it suggests that the non-elites would not protest when the income redistribution policy favors their interest, although it is possible to conduct a successful protest. This condition naturally reflects a realistic feature of the real world. For instance, Justino and Martorano (2019) provided evidence by contrapositive for our claim. They found that the willingness to participate in a protest is increasing along with the perceived low standards of living, suggesting that it is unlikely that a highly redistributive policy would trigger a protest.

[Figure 1 here]

In this fully defined condition, we could proceed to describe the payoff structure of the game explicitly. For the elites, it is always the case that perceived payoffs are more significant when the elites decide to conduct a clientelistic strategy than otherwise – recall that β_r, where $\beta_r < \bar{\beta}$, satisfies the condition that $\hat{y}^r(\tau = \tau^r) - \bar{\beta} y^r > \hat{y}^r(\tau = \tau^p)$. When the elites decided to isolate β_r fraction of their income in exchange for political support, the elites could set the tax rate at $\tau = \tau^r$ and, on the other hand, constrain the protest initiated by the non-elites, $\aleph^{\beth} < \aleph^p$; Hence, the elites' payoffs would be indifferent regardless of the non-elites actions (i.e., protest or accept), $U_3^r(\beta_r, \mathcal{P}) = U_4^p(\beta_r, \mathcal{A})$. While the elites' payoff is strictly smaller when they decide not to invest *de facto* power than otherwise. Under this circumstance, the elites' payoff would be more substantial when the non-elites decided to accept, $U_1^r(0, \mathcal{P}) < U_2^r(0, \mathcal{A})$, because the elites received nothing when the elites ran a protest. Besides, we define $U_2^r(0, \mathcal{A})$ as the elites' payoff under a stable democracy where *de facto* and *de jure* power are equally distributed among the citizens, there is no protest against it. In general, the payoff structure of the elites is defined as follows:

$$U_1^r(0, \mathcal{P}) < U_2^r(0, \mathcal{A}) < U_3^r(\beta_r, \mathcal{P}) = U_4^r(\beta_r, \mathcal{A}) \tag{19}$$

For the non-elites, the acceptance action would strictly dominate the protest for every choice node. Firstly, the most considerable payoff received by the non-elites occurred when the elites decided not to conduct a clientelistic strategy, and the non-elites accepted the situation. When the non-elites decide not to invest their *de facto* power through a clientelistic strategy, the payoff received by the non-elites is strictly higher when they choose to accept rather than otherwise. Similarly, when the elites decided to transfer β_r fraction of their income in exchange for political support, the non-elites payoff is also strictly higher when they choose to accept rather than otherwise where $\hat{y}^p(\tau = \tau^r) < \hat{y}^p(\tau = \tau^r) + \beta_r y^r/(1-\delta)$. Besides, we compare $U_2^p(0, \mathcal{A})$ and $U_4^p(\beta_r, \mathcal{A})$. Recall that since β_r ensures that the elites can sufficiently capture $\varphi_r(\beta_r)$ de facto power and set their preferred tax rate that optimizes their post-tax income (i.e., $\tau = \tau^r$) where $\hat{y}^r(\tau = \tau^r) > \hat{y}^r(\tau = \tau^p)$ in which also satisfies the condition that

$\hat{y}^r(\tau = \tau^r) - \overline{\beta} y^r > \hat{y}^r(\tau = \tau^p)$ where $\beta_r < \bar{\beta}$; therefore, $\hat{y}^r(\tau = \tau^r) - \beta_r y^r < \hat{y}^r(\tau = \tau^p)$ with $\bar{y}$ is constant. It implies that the non-elite payoff is more substantial under a stable democracy than in a captured democracy.

$$U_3^p(\beta_r, \mathcal{P}) < U_4^p(\beta_r, \mathcal{A}) \text{ and } U_1^p(0, \mathcal{P}) < U_2^p(0, \mathcal{A}) \tag{20}$$

Now we can see that, by applying backward induction, when the elites conduct clientelistic strategy at β_r, the non-elites would accept it, $\mathcal{A}$. Similarly, the non-elites payoff would be strictly better off when they decide to accept, $\mathcal{A}$, even when the elites do not conduct a clientelistic strategy. In this regard, therefore, $\mathcal{A}$ would strictly dominate $\mathcal{P}$ at every choice node of the non-elites. Given this structure, the elites thus strictly prefer to conduct a clientelistic strategy at β_r as it generates a higher payoff than otherwise. As such, the game has a unique subgame perfect equilibrium at $Z_4 = \left(U_4^r, U_4^p\right)$.

Equilibrium 1. Unique subgame perfect equilibrium is $S^* = (\beta_r, \mathcal{A})$.

4 The Shutdown of Clientelism

In this paper, we demonstrate that, under particular circumstances, democracy would virtually motivate political clientelism. The elites would rationally invest some portion of their income to the non-elites for political support. By this, they obtain de facto power and successfully reduce the threat of protest. It works when assumption 1 holds. That will not be the case when we relax this assumption. Assumption 1 ensures a high level of income inequality, which satisfies $\theta > \delta$ and $\theta > \theta^*$, allows the non-elites to conduct a clientelistic strategy, resulting in concentration control over economic policies and weakening the threat of protest.

Now consider that the initial level of income inequality, θ, is strictly lower than θ^* and satisfy $\theta > \delta$. In this setting, the comparative upper- and lower-bound of the elites' rational space is thus defined as follows:

$$\beta^* > \overline{\beta} \tag{21}$$

Equation (21) suggests that the elites' rational space's lower bound exceeds its upper bound, implying that they have no room to conduct a clientelistic strategy rationally, allowing for no political deviations from the elites. Under such circumstances, the game generates a unique subgame perfect equilibrium at Z_2. Adjacent to the cost-based view, it illustrates that as a country develops more equally, implying wealthier median voters, clientelism should naturally be eroded as a dominant form of political exchange because it extensively costs the ruling elites. Furthermore, it implies that, in line with the constrained-based perspective, when economic resources are not highly concentrated, employing the clientelistic strategy is unlikely to diminish the influence of protests. This would lead to a greater public sphere and a more effective mechanism for controlling political-business elites. In such a situation, the elites would opt not to engage in clientelism.

Equilibrium 2. Unique subgame perfect equilibrium is $S^* = (0, \mathcal{A})$.

Proof: When the elites conduct a clientelistic strategy at $\beta_r > \beta^*$, it possibly ensures a more significant political power and the weaker threat of the protest but does not necessarily provide higher benefit than otherwise, where $\hat{y}^r(\tau = \tau^r) - \overline{\beta} y^r < \hat{y}^r(\tau = \tau^p)$ or $\overline{\beta} > \tau^p$. On the one hand, when they decide to conduct a clientelistic strategy at $\beta_r < \overline{\beta}$, it is possibly enforceable but does not necessarily provide higher de facto political power and the manageable threat of protest, so, in other words, the elites invest for nothing. In this case, the elites would choose to conduct no political clientelism, $\beta = 0$

5 Empirical Strategy

5. 1 Measuring Clientelism

Empirical measurement of clientelistic schemes is a challenging effort. Despite its consensus about the definition, there is no one-size-fits-all measure for clientelism. Several works measured clientelism based on a survey. Berenschot (2018) developed an index, the so-called Clientelism Perception Index (CPI), to investigate the clientelism phenomenon in Indonesia based on a massive survey capturing seven aspects of clientelism (i.e., governmental contracts and jobs, public services, access to social welfare, use of social assistance funds, paperwork, and money politics). This approach results in a comprehensive measurement of clientelism. Based on a survey, Papadopoulos (1997) investigated the transformation of party clientelism behavior in Southern Europe.

Several empirical works used proxy variables to capture the clientelistic scheme as an alternative to the survey-based measure, allowing for more extensive panel analysis. For example, Keefer (2007) conducted a cross-country investigation and used patronage spending as the proxy variable. Consistent with the definition, he argued that clientelistic politics should necessarily be associated with relatively high patronage spending. In this regard, the ruling elites promise some benefits, including jobs in the government sector (Keefer 2007) and particularly influential people to mobilize votes (Gottlieb and Larreguy 2020). Following Keefer (2007), we employ the share of employee compensation and wage and salaries to public spending to measure patronage clientelism. These variables could portray targeted benefits to particular constituencies pervasive in clientelist democracy (Robinson and Verdier 2013; Keefer 2007).

5. 2 Econometric Method

In this paper, we consider the following a simple econometric specification as follows:

$$\mathcal{C}_{it} = \alpha + \beta\theta_{it} + \varphi' W + \varepsilon_{it} \qquad (22)$$

where $\mathcal{C}$ denotes the proxy variables of clientelism as the dependent variable, θ is income inequality as independent variables, and W for the set of control variables.

We include several control variables representing demographic, cultural, and economic factors to avoid model misspecification bias. Following Keefer (2007), we employ total population and ethnic fractionalization for demographic and cultural factors and share of agriculture and mining sectors to GDP for the economic variables that are expected to capture a so-called resource curse (e.g., see Robinson, Torvik, and Verdier, 2006).

Estimating equation (22) using Ordinary Least Square (OLS) would be problematic as it potentially suffers from an endogeneity problem. Moreover, intuitively, the Gini coefficient is impacted by other related variables. In this regard, we require using instrumental variable methods, such as Two-stage Least Square (2SLS), to obtain consistent estimation (Baltagi 2005). For example, consider the following structural econometric specification as follows:

$$y_1 = Z_1\delta_1 + u_1 \tag{23}$$

where u_1 is considered as a one-way error composed that is $u_1 = Z_\mu \mu_1 + v_1$ with random vectors by zero means and covariance. $Z_1 = [Y_1, X_1]$ and $\delta_1' = (\gamma_1', \beta_1')$. More specifically, Y_1 and X_1 respectively denote the set of g_1 endogenous variables and the set of k_1 included exogenous variables. Now let $X = [X_1, X_2]$ be the set of all exogenous variables in the system so that such an equation is identified with k_2 where the number of excluded exogenous variables is weakly larger than g_1.

There are several ways to estimate equation (23), but in general, (Baltagi 2005) suggested three main econometric approaches: Fixed Effects Two-stage Least Square (FE2SLS), Between Two-stage Least Square (B2SLS), and Error Component Two-stage Least Square (EC2SLS). Contrary to FE2SLS, the B2SLS estimates control no cross-sectional effects, resulting in higher deterrent effects than the fixed effects estimates. On the other hand, EC2SLS deals with the endogeneity problem using a random-effects 2SLS estimator with a matrix-weighted average of B2SLS and FE2SLS. In selecting the most-fitted estimator for IV-based panel data regression, (Baltagi 2005) suggested that it is essential to see the *f-statistic value* reporting whether or not cross-sectional effects from F2SLS estimation are equal to zero, $\mu_i = 0$. If the probability of *f-statistic* is below (above) the confidence level, the null hypothesis of $\mu_i = 0$ should be rejected (accepted). It implies that eliminating μ_i would (not) adversely affect the estimation so that the F2SLS estimates better (worse) than B2SLS. Second, we used the Hausman Test to compare the selected model from the f-test to EC2SLS. If the Hausman test does not reject the null hypothesis, it implies that EC2SLS outperforms either FE2SLS or B2SLS.

Lastly, we use the theoretical basis of income inequality determinants in choosing the instrumental variables. First, we use corruption to capture the institutional factor of inequality. (Kunawotor, Bokpin, and Barnor 2020; Apergis, Dincer, and Payne 2010; Dincer and Gunalp 2012) provided empirical evidence concerning the relationship between corruption and inequality. Second, we use income per capita (and its squared value) to capture Kuznets' inverted U-curve hypothesis. This variable is extensively utilized in the literature (see Acemoglu, Johnson, Robinson, and Yared, 2008; Acemoglu et al., 2015; Lee, 2005). Third, the degree of democracy. It precisely captures the moderating effect of democracy on income inequality (Acemoglu et al., 2008; Acemoglu et al., 2015; Lee, 2005; Meltzer and Richard, 1983).

5. 3 Data and Variables

This paper constructs an unbalanced panel dataset of 104 democratic countries from 2002 to 2017. The observation period is selected based on the availability of the dataset. We use the Polity IV index as the yardstick for the cross-sectional observations in selecting the democratic countries. Countries with a democratic regime should satisfy the value of the Polity IV index above six, while otherwise, for a non-democratic regime. Table 1 displays the descriptive statistics of our variables.

Table 1 here

For the dependent variable, we collect the data for Employee Compensation (% of Public Expense) and Wage and Salaries (% of Public Expense) from Government Finance Statistics (IMF) consistent with Keefer (2007). These variables represent the government wage bills on government employees. For the income inequality variable, as the dependent variable, we use the Standardized World Inequality Indicators Database (SWIID) constructed by Solt (2020). Using SWIID is advantageous because it provides the gross Gini coefficients and the net Gini after taxes and transfers concurrently (Acemoglu et al., 2015). In this regard, we specifically utilize the net Gini coefficient as it aligns with our theoretical basis. Finally, in defining democracy, we use Political Regime (Polity IV). Technically, the higher degree of democracy is characterized by the higher value of the Political Regime, where values above six indicate a democratic regime.

Conversely, values below six suggest a non-democratic regime. For the share of agriculture and mining, population (in millions), secondary school enrollment, education expenditure, and income per capita, we collect these variables from the Arbetman-Rabinowitz et al. (2011) dataset. For the Ethnic Fractionalization variable, we collect data from the Historical Index of Ethnic Fractionalization (HIEF) dataset (Drazanova 2019), by which Alesina, Devleeschauwer, Easterly, and Kurlat (2003) developed the method. Lastly, we use the education budget as an additional control variable in estimating the clientelism which is obtained from (Arbetman-Rabinowitz et al. 2011).

6 Results

The results are exhibited in Table 2, presenting the empirical relationship between patronage and income inequality. For the estimation using employee compensation as the dependent variable, the *f-statistic* is 80.31 with a probability below one percent confidence level. It indicates that the unobserved cross-sectional components, μ_i, are significantly different from zero, so we could not reject that FE2SLS estimation is better than B2SLS. For the Hausman test, comparing FE2SLS and EC2SLS, the value of $\chi^2(5)$ is 8.84 with a probability of 0.1155, suggesting that we could not reject that EC2SLS outperforms FE2SLS. For the second model estimate, the *f-statistic* is 114.44, with the probability of being below the one percent confidence level. We should reject the null hypothesis, implying that B2SLS is better than FE2SLS. The Hausman test, $\chi^2(5)$ is 8.55 with a probability of 0.1285, implying that we could not reject the null hypothesis that EC2SLS is consistent and efficient. Based on these results, therefore, we estimate both specifications using EC2SLS. The Sargan-Hansen test, inspecting the overidentifying restrictions of the instrumental variables, is insignificant for both specifications. It implies that the instrumental variables as a group are exogenous.

[Table 2 here]

Our estimations demonstrate that income inequality significantly affects the conduct of patronage clientelism in the way we expect. Rising income inequality by one percent significantly increases the portion of government employee compensation by 0.008 percent. For clientelism in the form of wage and salaries of government employees, our estimation suggests that a one percent increase in income inequality would significantly increase wage and salary expenditure up to 0.011 percent. These findings confirm our theoretical predictions that a high concentration of wealth, implying a lower income of people experiencing poverty, allows the elites to conduct political clientelism because it is enforceable and effective to capture sufficiently de facto and collective action power. In the case of patronage clientelism, the ruling elites would provide some fraction of their budget to enlarge the bureaucratic body in exchange for political support as income inequality rises. The collective-action problem then arises as the ruling elites or politicians could offer personal benefits to the leaders of politically influential groups through a job at the office (Gottlieb and Larreguy, 2020). Such a strategy would be effective and enforceable because poor voters are risk-averse and, hence, more easily tempted by clientelistic benefits (Brusco, Nazareno, and Stokes, 2004; Scott, 1972). On the other hand, as a country develops more economically equal, implying wealthier bottom-income voters, clientelism should naturally be eroded. This is because it extensively costs the ruling elites to provide a relatively higher wage compensation than the citizens' income.

7 Robustness Exercises

This section performs robustness exercises to check whether our findings are consistent throughout alternative empirical strategies. For the first robustness exercise, we use the level of secondary school enrollment, representing the degree of non-targeted policy as the dependent variable. For example, Keefer (2007) argued that clientelistic practices would be reflected by relatively high patronage employment and an inadequate provision of non-targeted public goods, especially in education. He also suggested that secondary school enrollment is likely to measure the better-perceived quality of education to all children than public education spending.

[Table 3 here]

Table 3 displays the inequality and clientelism nexus results in non-targeted policy, i.e., secondary school enrollment. Although income inequality is statistically significant only in EC2SLS estimation, the sign of parameter obtained from these estimations suggests a consistent result to our theoretical conjecture. Income inequality makes the non-targeted policy less pervasive. People with low incomes have relatively weak political power, so the ruling elites would necessarily conduct electoral targeting. On the other hand, given the trade-off between the share of non- and targeted public goods, the politicians tend to provide short-term assistance to some fraction of people experiencing poverty, not long-term forms of public goods like education, when income inequality is sufficiently high. The rationale is that the poor live daily, preventing far-sighted objectives, unlike the rich. In other words, people experiencing poverty would likely prefer short-term assistance such as transfers, which then causes the ruling elites to provide such transfers to gain political support. Consequently, it results in diminishing non-targeted public goods.

[Table 4 here]

We consider non-instrumental variable panel data regression estimations for the second robustness exercise. We perform Pooled Least Squared (PLS) with country-clustered robust standard error to control the serial correlation of unobserved variables. This exercise captures whether our findings would be consistent when disregarding the endogeneity problem. Although the estimations might suffer from several econometrical problems, we must check our findings' consistency based on alternative empirical assumptions. Our results are exhibited in Table 4 and demonstrate that income inequality significantly affects clientelism practices in patronage employment and non-targeted policy. Furthermore, the sign of estimated parameters is consistent with our theoretical conjecture and previous estimations in Tables 3 and 4, indicating that the empirical results are robust.

8 Conclusion

A growing literature on the political economy of clientelism considers cost and constraint perspectives differently. In this paper, we argue that these perspectives are neither inconsistent nor refuting, while they virtually provide a holistic view of the political economy of clientelism. The cost-based perspective gives an intuition about the cost feasibility of political clientelism under particular economic conditions. The constraint-based perspective provides the importance of collective action in curtailing political clientelism.

In this paper, we conduct game-theoretical modeling to illustrate the making of political clientelism by synthesizing cost and constraint perspectives. Our model illustrates that when income inequality is sufficiently high, it is rational for the elites to conduct political clientelism as a stable democracy would extensively cost them. In this case, the elites would invest some portion of their income to increase de facto power and capture the power of protest by providing private benefits for some fraction of the non-elites in exchange for political support. On the contrary, clientelistic strategies would be irrational when income inequality is sufficiently low. As a country develops more equally, clientelism should naturally be eroded as it extensively costs the ruling elites and is ineffective in eroding the power of protest. Therefore, the elites would rationally conduct no clientelistic strategy.

Our empirical results confirm the theoretical conjecture suggesting that clientelism is robustly related to income inequality by utilizing two-stage least squared estimations. When income inequality rises, political clientelism will be more pervasive. A high concentration of wealth, implying a lower income of people experiencing poverty, allows the elites to conduct political clientelism because it is enforceable and effective to capture sufficiently de facto and collective action power. Brusco, Nazareno, and Stokes (2004) and Scott (1972) argued that a clientelistic strategy would be more effective as poor voters are risk-averse and more easily tempted by clientelistic benefits.

Compliance with Ethical Standards:
The authors declare that they have no conflict of interest.
This article contains no studies with human participants or animals performed by any of the authors.

References

Acemoglu, By Daron, Simon Johnson, James A Robinson, and Pierre Yared. 2008. "Income and Democracy," no. 1959: 808–42.

Acemoglu, Daron, Suresh Naidu, Pascual Restrepo, and James A. Robinson. 2015. *Democracy, Redistribution, and Inequality*. *Handbook of Income Distribution*. 1st ed. Vol. 2. Elsevier B.V. https://doi.org/10.1016/B978-0-444-59429-7.00022-4.

Acemoglu, Daron, and James A. Robinson. 2000. "Why Did the West Extend the Franchise? Democracy, Inequality, and Growth in Historical Perspective." *Quarterly Journal of Economics* 115 (4): 1167–99. https://doi.org/10.1162/003355300555042.

———. 2006. *Economic Origins of Dictatorship and Democracy*. Cambridge CB2 2RU,UK: Cambridge University Press.

———. 2008. "Persistence of Power, Elites, and Institutions." *American Economic Review* 98 (1): 267–93. https://doi.org/10.1257/aer.98.1.267.

Alesina, Alberto, Arnaud Devleeschauwer, William Easterly, and Sergio Kurlat. 2003. "Fractionalization." *Journal of Economic Growth* 8: 155–94.

Apergis, Nicholas, Oguzhan C. Dincer, and James E. Payne. 2010. "The Relationship between Corruption and Income Inequality in U.S. States: Evidence from a Panel Cointegration and Error Correction Model." *Public Choice* 145 (1): 125–35. https://doi.org/10.1007/s11127-009-9557-1.

Arbetman-Rabinowitz, Marina, Ali Fisunoglu, Jacek Kugler, Mark Abdollahian, Kristin Johnson, and Kyungkook Kang. 2011. "Replication Data for: Relative Political Capacity Dataset." Harvard Dataverse, V4. https://doi.org/https://doi.org/10.7910/DVN/NRR7MB.

Aspinall, Edward. 2013. "A Nation in Fragments: Patronage and Neoliberalism in Contemporary Indonesia." *Critical Asian Studies* 45 (1): 27–54. https://doi.org/10.1080/14672715.2013.758820.

Aspinall, Edward, and Ward Berenschot. 2019. "Campaign Financing, Business, and the Public Sphere." In *Democracy for Sale: Election, Clientilism, and the State in Indonesia*. Cornell University Press.

Baltagi, Badi H. 2005. *Econometric Analysis of Panel Data*. 3rd Editio. West Sussex PO19

8SQ, England: John Wiley & Sons Ltd.
Barbera, Salvador, and Matthew Jackson. 2016. “A Model of Protests, Revolution, and Information.” *Quarterly Journal of Political Science* 15 (3): 297–335. https://doi.org/10.1561/100.00019002.
Bellinger, Paul T., and Moisés Arce. 2011. “Protest and Democracy in Latin America’s Market Era.” *Political Research Quarterly* 64 (3): 688–704. https://doi.org/10.1177/1065912910373557.
Berenschot, Ward. 2018. “The Political Economy of Clientelism: A Comparative Study of Indonesia’s Patronage Democracy.” *Comparative Political Studies* 51 (12): 1563–93. https://doi.org/10.1177/0010414018758756.
Brusco, Valeria, Marcelo Nazareno, and Susan C. Stokes. 2004. “Vote Buying in Argentina.” *Latin American Research Review* 39 (2): 66–88. https://doi.org/10.1353/lar.2004.0022.
Chandra, Kachan. 2004. *Why Ethnic Parties Succeed: Patronage and Ethnic Head Counts in India*. Cambridge, UK: Cambridge University Press.
Dincer, Oguzhan C., and Burak Gunalp. 2012. “Corruption and Income Inequality in the United States.” *Contemporary Economic Policy* 30 (2): 283–92. https://doi.org/10.1111/j.1465-7287.2011.00262.x.
Drazanova, Lenka. 2019. “Historical Index of Ethnic Fractionalization Dataset (HIEF).” Harvard Dataverse. https://doi.org/doi:10.7910/DVN/4JQRCL.
Gottlieb, Jessica, and Horacio Larreguy. 2020. “An Informational Theory of Electoral Targeting in Young Clientelistic Democracies: Evidence from Senegal.” *Quarterly Journal of Political Science* 15 (1): 73–104. https://doi.org/10.1561/100.00019018.
Justino, Patricia, and Bruno Martorano. 2019. “Redistributive Preferences and Protests in Latin America.” *Journal of Conflict Resolution* 63 (9): 2128–54. https://doi.org/10.1177/0022002719827370.
Keefer, Philip. 2007. “Clientelism, Credibility, and the Policy Choices of Young Democracies.” *American Journal of Political Science* 51 (4): 804–21. https://doi.org/10.1111/j.1540-5907.2007.00282.x.
Kunawotor, Mark Edem, Godfred Alufar Bokpin, and Charles Barnor. 2020. “Drivers of Income Inequality in Africa: Does Institutional Quality Matter?” *African Development Review*, no. 2015: 1–12. https://doi.org/10.1111/1467-8268.12473.
Kurtz, Marcus J. 2004. “The Dilemmas of Democracy in the Open Economy: Lessons from Latin America.” *World Politics* 56 (2): 262–302. https://doi.org/10.1353/wp.2004.0013.
Lee, Cheol Sung. 2005. “Income Inequality, Democracy, and Public Sector Size.” *American Sociological Review* 70 (1): 158–81. https://doi.org/10.1177/000312240507000108.
Magolini, Beatriz, Alberto Diaz-Cayeros, and Fererico Estévez. 2007. “Clientelism and Portfolio Diversification: A Model of Electoral Investment with Applications to Mexico.” In *Patrons, Clients, and Policies*, edited by H. Kitschelt and S. I. Wilkinson, 182–205. Cambridge University Press.
Meltzer, Allan H., and Scott F. Richard. 1983. “A Rational Theory of the Size of Government.” *Public Choice* 41 (3): 403–18. https://doi.org/10.1007/BF00141072.
Moseley, Mason W. 2015. “Contentious Engagement: Understanding Protest Participation in Latin American Democracies.” *Journal of Politics in Latin America* 7 (3): 3–48. https://doi.org/10.1177/1866802x1500700301.

Oktariza, Noory. 2020. *Democracy for Sale: Elections, Clientelism, and the State of in Indonesia*. *Bulletin of Indonesian Economic Studies*. Vol. 56. https://doi.org/10.1080/00074918.2020.1742957.

Papadopoulos, Yannis. 1997. "Transformations of Party Clientelism in Southern Europe in a Phase of Democratic Consolidation." *Swiss Political Science Review* 3 (4): 1–9. https://doi.org/10.1002/j.1662-6370.1997.tb00226.x.

Robinson, James A., Ragnar Torvik, and Thierry Verdier. 2006. "Political Foundations of the Resource Curse." *Journal of Development Economics* 79 (2): 447–68. https://doi.org/10.1016/j.jdeveco.2006.01.008.

Robinson, James A., and Thierry Verdier. 2013. "The Political Economy of Clientelism." *Scandinavian Journal of Economics* 115 (2): 260–91. https://doi.org/10.1111/sjoe.12010.

Scott, James C. 1972. "The Erosion of Patron-Client Bonds and Social Change in Rural Southeast Asia." *The Journal of Asian Studies* 32 (1): 5–37. https://doi.org/10.2307/2053176.

Solt, Frederick. 2020. "Measuring Income Inequality Across Countries and Over Time: The Standardized World Income Inequality Database." *Social Science Quarterly* 101 (3): 1183–99. https://doi.org/10.1111/ssqu.12795.

Appendix A
Proof of Proposition 3

In this section, we illustrate the construction of how β determines both the collective-action problems and the outcomes of non-elites actions. We assume that both protesters and non-protesters use exclusion strategy to prevent the free-riding considerations. Suppose that the elites provide $\beta_{\Xi} \leq \beta^*$ which leads to $\aleph^{\Xi} \geq \aleph^p$; hence, it brings success in protest. In this regard, the non-elites who participate (i.e., protesters) in the protest would receive the post-protest income as follows:

$$y^p + \frac{(1-\mu)y^r}{\aleph^{\Xi}} - \varepsilon\bar{y}$$

Where $y^p(\tau = \tau^r)$ is the post-tax income of non-elites with $\tau = \tau^r = 0$; $(1-\mu)y^r/\aleph^{\Xi}$ is defined as the additional income perceived by the protesters from imposing the elites to redistribute some fraction of their income; and $\varepsilon\bar{y}$ is a portion of the economy damaged by the protest.

On the other hand, the non-protesters would receive the following payoff if they do not join the protest:

$$y^p + \frac{\beta_{\Xi} y^r}{(1-\delta) - \aleph^{\Xi}}$$

Where $\beta_{\Xi} y^r/(1-\delta) - \aleph^{\Xi}$ is additional income received by the non-protesters in which transferred from the elites.

By relaxing that the elites invest $\beta_{\Xi} = 0$ and thus $\aleph^{\Xi} \geq \aleph^p$ still holds, we could easily compare the benefit of joining the protest as follows:

$$y^p + \frac{(1-\mu)y^r}{\aleph^{\Xi}} - \varepsilon\bar{y} > y^p \text{ for } \frac{(1-\mu)y^r}{\aleph^{\Xi}} > \varepsilon\bar{y} \tag{24A.1}$$

We now consider the opposite case in which the elites invest $\beta_{\beth} > \beta^*$ that leads to an insufficient number of protesters, $\aleph^{\beth} < \aleph^p$; therefore, the protest fails. In this case, the protest would neither cost the economy, $\varepsilon\bar{y} = 0$, nor receive the expropriation benefits from the elites, $(1-\mu)y^r/\aleph^{\beth} = 0$. Therefore, the protesters would receive simply y^p. On the other hand, the non-protesters would receive an extra income from the elites by $\beta_{\beth} y^r/(1-\delta) - \aleph^{\beth} > 0$. In this regard, therefore, the payoff's comparison between joining and refuting the protest is defined as follows:

$$y^p + \frac{\beta_{\beth} y^r}{(1-\delta) - \aleph^{\beth}} > y^p \Rightarrow \frac{\beta_{\beth} y^r}{(1-\delta) - \aleph^{\beth}} > 0 \tag{A.2}$$

Equations (A.1) and (A.2) rationalize that the non-elites payoffs of either joining or refusing the protest are dependent on the level of political clientelism. When the elites conduct an inadequate clientelistic strategy, it leads to a higher payoff of joining the protest. On the contrary, the non-elites would be better off by refusing the protest when the elites transfer sufficient money.

Appendix B

Income Inequality and Patronage Clientelism: FE2SLS and B2SLS Estimations

	Employment Compensation		Wage and Salaries	
	(1)	(2)	(1)	(2)
Income Inequality	0.0107487***	0.0041196*	0.0023795	0.0063714**
	(0.0037835)	(0.0022402)	(0.0042133)	(0.0028049)
Population	0.0000326	-0.0001979	0.0005661	-0.0002603
	(0.0005371)	(0.000148)	(0.0008561)	(0.0002077)
Ethnic Fractionalization	-0.2854615***	-0.012379	-0.0807147	-0.0811689
	(0.0689264)	(0.0534626)	(0.0791144)	(0.0771901)
Share of Agriculture	0.0889632	0.5652591***	-0.1051204	0.6212316***
	(0.0882605)	(0.2102691)	(0.0815522)	(0.2396019)
Share of Mining	0.0070904	-0.1252052	-0.1017728	-0.066004
	(0.0878058)	(0.1828421)	(0.0818328)	(0.2041281)
Constant	-0.0029995	0.1201509**	0.1788641	0.0076391
	(0.1308449)	(0.0597778)	(0.1377515)	(0.0751256)
R-squared	0.3415	0.4776	0.0482	0.6079
Observations	603	603	409	409
Number of Countries	60	60	45	45
Estimator	FE2SLS	B2SLS	FE2SLS	B2SLS

Notes: The asterisk denotes statistical significance *, **, and *** at 10 percent, 5 percent, and 1 percent, respectively. Numbers in the parentheses ( ) represent standard error.

Figure 1
The Extensive Form of Political Clientelism Game

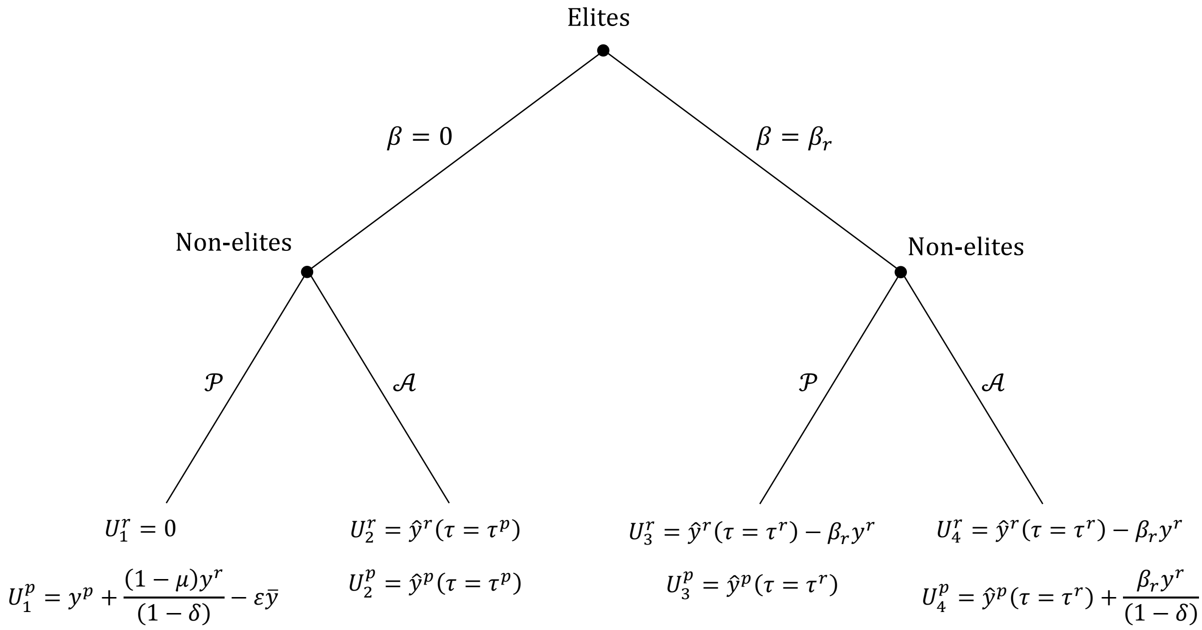

Table 1
Descriptive Statistics

Variable	Observations	Mean	Std. Dev.	Min	Max
Gini Coefficient	1,347	38.12	9.19	23.00	66.90
Control of Corruption	1,347	0.33	1.01	(1.50)	2.47
Ethnic Fractionalization	1,015	0.41	0.24	0.02	0.89
Tax to GDP Ratio	1,041	0.21	0.07	0.06	0.61
Share of Agriculture Sector	1,050	0.11	0.12	0.00	0.70
Share of Mining Sector	1,050	0.03	0.06	0.00	0.32
Income per Capita	1,050	14,869.18	14,372.57	372.60	80,215.48
Secondary School Enrollment	886	0.8628	0.2576	0.0894	1.6558
Public Expenditure on Education	663	5.0433	5.1890	0.0189	19.9275
Population	1,347	41.85	130.56	0.44	1,310.15
Public Budget on Education	663	504.33	518.91	1.90	1,992.75
Employee Compensation	836	0.28	0.08	0.14	0.59
Wage and Salaries	583	0.24	0.10	0.12	0.59

Table 2
Inequality and Clientelism

	Employee Compensation	Wage and Salaries
Income Inequality	0.0087342***	0.0115581***
	(0.0019575)	(0.0025964)
Population	-0.0003029*	-0.0003113
	(0.0001575)	(0.0003298)
Ethnic Fractionalization	-0.1521566**	-0.2107228***
	(0.0640499)	(0.0797328)
Share of Agriculture	0.1019852	-0.0387497
	(0.0947235)	(0.1146267)
Share of Mining	-0.0362015	0.0030847
	(0.0928399)	(0.1121003)
Constant	0.0326161	-0.0944693
	(0.0534072)	(0.0720983)
Wald χ^2	52.46***	35.78***
R-squared	0.4316	0.5333
F-statistic	80.31***	114.44***
Hausman Test	8.84	8.55
Sargan-Hansen Test	13.02	14.54
Observations	603	409
Number of Countries	60	45
Estimator	EC2SLS	EC2SLS

Notes: The asterisk denotes statistical significance *, **, and *** at 10 percent, 5 percent, and 1 percent, respectively. Numbers in the parentheses ( ) represent clustered-robust standard error. F-statistic test obtained from FE2SLS estimation with no robust standard error, testing the null hypothesis of $\mu_i = 0$ (see FE2SLS and B2SLS estimations in Appendix B). Hausman Test is conducted to compare FE2SLS to EC2SLS where EC2SLS is consistent under the null hypothesis. Sargan-Hansen Test examines the overidentifying restrictions on Instrumental Variables. The endogenous variable is income inequality with the following excluded instrumental variables: Control of corruption, democracy index (Polity IV), tax to GDP ratio, income per capita, and squared value of income per capita.

Table 3
Non-targeted Policy and Income Inequality

	(1)	(2)	(3)
Income Inequality	-0.0609702	-0.0141154***	-0.0050323
	(0.0492196)	(0.004553)	(0.0066156)
Population	0.0077389	-0.0001915	-0.0004401**
	(0.0066116)	(0.0002206)	(0.0002241)
Ethnic Fractionalization	0.5107023	0.095961	0.0240265
	(0.4573754)	(0.1134967)	(0.1412623)
Share of Agriculture	-0.400203	-1.381392***	-1.671453***
	(0.6896278)	(0.2891059)	(0.2521259)
Share of Mining	-0.2935877	0.1300963	-0.2080045
	(0.6274122)	(0.1911245)	(0.4057791)
Education Expenditure	-0.0037064	-0.0000639	0.0104492*
	(0.0029918)	(0.0019875)	(0.0061792)
Constant	2.695929	1.474211	1.165733***
	(1.564682)	(0.1352809)	(0.2186705)
Wald Chi-squared	6.99	137.25	210.87
R-squared	0.1638	0.599	0.6297
Sargan-Hansen Statistic	7.067	24.003**	5.146
Observations	578	578	578
Number of Countries	74	74	74
Estimator	FE2SLS	EC2SLS	B2SLS

Notes: The asterisk denotes statistical significance *, **, and *** at 10 percent, 5 percent, and 1 percent, respectively. Numbers in the parentheses ( ) represent clustered-robust standard error. Sargan-Hansen Test inspects the overidentifying restrictions on Instrumental Variables. The endogenous variable is income inequality with the following excluded instrumental variables: Control of corruption, democracy index (Polity IV), tax to GDP ratio, income per capita, and squared value of income per capita.

Table 4
Income Inequality and Clientelism: Pooled Least Squared Estimations

	Employee Compensation	Wage and Salaries	Secondary School Enrollment
Income Inequality	0.0053593***	0.0073626***	-0.0081531***
	(0.0012317)	(0.0015205)	(0.0022331)
Population	-0.000274*	-0.0003664	-0.0002959**
	(0.0001445)	(0.0002353)	(0.0001294)
Ethnic Fractionalization	-0.0308686	-0.0756334	0.0935525
	(0.0443706)	(0.0588762)	(0.0827984)
Share of Agriculture	0.4133923**	0.4839609***	-1.640754***
	(0.1666372)	(0.1999535)	(0.2430083)
Share of Mining	-0.0908904	-0.0226581	0.1476939
	(0.2022649)	(0.1891431)	(0.2823723)
Education Expenditure			0.0074284**
			(0.0033158)
Constant	0.0952196***	-0.0156435	1.253307***
	(0.0324874)	(0.0459102)	(0.0729592)
F-statistic	9.14***	12.57***	34.02***
R-squared	0.4903	0.616	0.6415
Observations	603	409	578
Number of Countries	60	45	74

Notes: The asterisk denotes statistical significance *, **, and *** at 10 percent, 5 percent, and 1 percent, respectively. Numbers in the parentheses ( ) represent clustered-robust standard error